MILITARY SPECIFICATION SHEET

ELECTRON TUBES, RECEIVING

TYPES 6252 AND 6907

The complete requirements for procuring the electron tubes described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Push-pull rf beam power amplifiers (see note 1)
F1 = 250 MHz (6252 and 6907); F2 = 600 MHz (6907)

Outline --- See figure 1
Cathode --- Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	Top leads
Element	---	h	2g1	g2	k int sd	hct	1g1	h	1a, 2a

ABSOLUTE MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ecc2	(A) Ehk	Ib/a
Unit:	V	Vdc	Vdc	Vdc	Vdc	v	mAdc
Maximum:							
Class C, Telep:	6.9 or 13.8	500	-200	250	600	100	60
Class C, Teleg:	6.9 or 13.8	600	-200	300	600	100	75
Minimum:	5.7 or 11.4	---	---	---	---	---	---
TEST CONDITIONS:	6.3	300	---	250	---	---	40

ABSOLUTE MAXIMUM RATINGS:

Parameter:	Ic1/g	Pg1/g	Pg2/g	Pp/a	Pi/a	T	tk
Unit:	mAdc	W	W	W	W	°C	sec(min)
Maximum:							
Class C, Telep:	2.0	0.5	1.5	6.7	25	Note 2	---
Class C, Teleg:	2.5	0.5	2.0	10	36	Note 2	---
Minimum:	---	---	---	---	---	---	---
TEST CONDITIONS:	---	---	---	---	---	---	120

GENERAL:

Qualification - Required

(A) denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Qualification inspection							
1031	Low-frequency vibration(1)	F = 50 Hz; double amplitude = 0.080 inch; t= 900; no voltages	---	---	---	---	---	---
1236	Power oscillation (1) Type 6907	Class C amplifier; F = 600 MHz, push-pull operation; Eb = 400 Vdc; Ib = 100 mAdc; Ec2 = 250 Vdc (max); Rg = 0.0125 to 0.0167 Meg; Ic1 = 3 mAdc (see note 3)	---	---	Po	15	---	W
	Quality conformance inspection, part 1							
1231	Peak emission	eb = ec1 = ec2 = 225 v (see note 4)	0.65	II	is	1.0	---	a
1256	Electrode current (screen)	See note 4	0.65	II	Ic2	0	8.0	mAdc
1261	Electrode voltage (control grid)	See note 4	0.65	II	Ec1	-12.0	-20.0	Vdc
1266	Total grid current	t = 180 (see notes 2 and 4)	0.65	II	Ic1	---	-2.0	μAdc
1301	Heater current	t = 300	0.65	II	If	1.10	1.50	Adc
1201	Short and discontinuity detection		0.65	II	---	---	---	---
	Quality conformance inspection, part 2							
1031	Low-frequency vibration(2)	Ebb = 250 Vdc; Ec2 = 170 Vdc; Ec1/Ib = 10 mA; Rp = 2,000 ohms (see note 4)	---	---	Ep	---	500	mVac
1036	Bump	450 G	---	---	---	---	---	---
1331	Direct-interelectrode capacitance	See note 6	---	---	Cgp	---	0.07	pF
		See note 6			Cin	6.0	7.0	pF
		See note 6			Cout	2.0	3.0	pF
	Type 6252	See note 7	---	---	Cn	0.000	0.075	pF
	Type 6252	See note 7			ΔCn	-0.030	+0.030	pF
1336	Heater-cathode leakage		---	---	Ihk	---	20	μAdc
1236	Power oscillation (2)	Class C amplifier, push-pull operation F = 250 MHz; Eb = 600 Vdc; Ib = 100 mAdc; Ec2 = 250 Vdc (max); Rg1 = 0.02 to 0.03 Meg; Ic1 = 3 mAdc (see note 3)	1.0	II	Po	33	---	W
			---	---	Ic2	3.0	15.0	mAdc
1236	Power oscillation (3)	Power oscillation (2), except Ef = 5.7 V	---	---	ΔPo Ef	---	10	%

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued							
1236	Power oscillation (4) Type 6907	Class C amplifier, push-pull operation; F = 450 to 470 MHz; Ib = 115 mAdc; Eb = 425 Vdc; Ec2 = 250 Vdc (max); Rg1 = 0.0237 Meg; Ic1 = 2.0 to 3.0 mAdc (see note 3)	---	---	Po	15	---	W
(A)1211	Insulation of electrodes		4.0	S3	---	---	---	---
(A)1105	Permanence of marking		---	---	---	---	---	---
	Quality conformance inspection, part 3							
---	Life-test provisions	Group D; power oscillation (2)	---	---	---	---	---	---
---	Life-test end points (500 hours)	Total grid current	---	---	Ic1	---	-5.0	μAdc
		Peak emission	---	---	is	0.7	---	A
		Power oscillation (2)	---	---	Po	30	---	W

NOTES:

1. Tube type 6252 is provided with additional internal neutralizing capacitances. Type 6907 is not provided with additional internal neutralizing capacitances. For details see direct interelectrode capacitance (method 1331) and note 7.
2. Temperature of bottom seal - maximum 180°C.
   Temperature of anode seal - maximum 250°C.
   Temperature of bulb - maximum 250°C.
3. Power output is total power at drive frequency delivered to load. Drive power shall not be subtracted from total output.
4. Each unit shall be read separately. Control grid of unit not under test shall be connected to -100 Vdc.
5. This test to be performed at the conclusion of the holding period.
6. The test shall be made on each unit separately. The unit not under test shall be tied to ground. Shielding shall be in accordance with figure 2.
7. Cross neutralizing capacitance (Cn1) of unit 1 is defined as the capacitance between the control grid of unit 1 and the anode of unit 2. All elements not under test are grounded. Cn2 of unit 2 is defined as the capacitance between the control grid of unit 2 to the anode of unit 1 with all elements not under test grounded. ΔCn1 is defined as Cgp (unit 1)-Cn1. ΔCn2 is defined as Cgp (unit 2)-Cn2.

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Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army -
Navy -
Air Force - 11, 17
DSA - ES

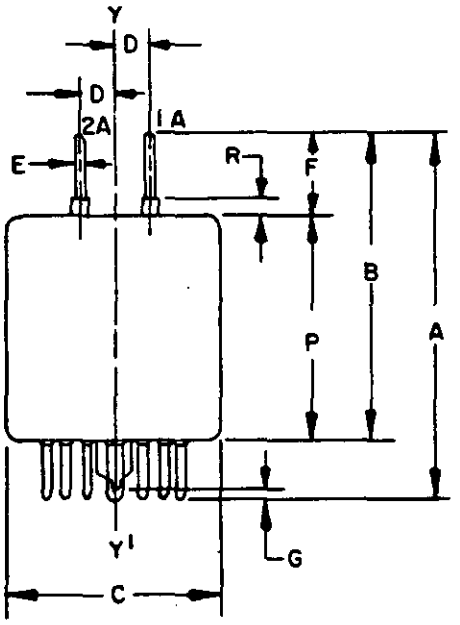
User activities:
Army - WC
Navy - AS, OS, CG, SH
Air Force - 19

Preparing activity: Navy - EC

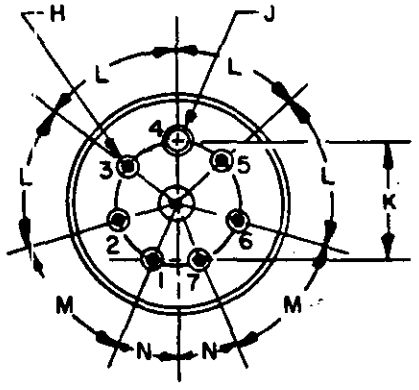
Agent: DSA - ES

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1		
A		3.375 (85.73)
C		1.811 (46.00)
Quality conformance inspection, part 2		
B		2.906 (73.81)
D	See note c	
E	.077 (1.96)	.083 (2.11)
F		.718 (18.24)
G	0.0	
H	.052 (1.32)	.060 (1.52)
J	.122 (3.10)	.128 (3.25)
K	See note b	
L	51°	
M	52°	
N	26°	
P	2.062 (52.37)	2.188 (55.58)
R		.125 (3.18)



NOTES:

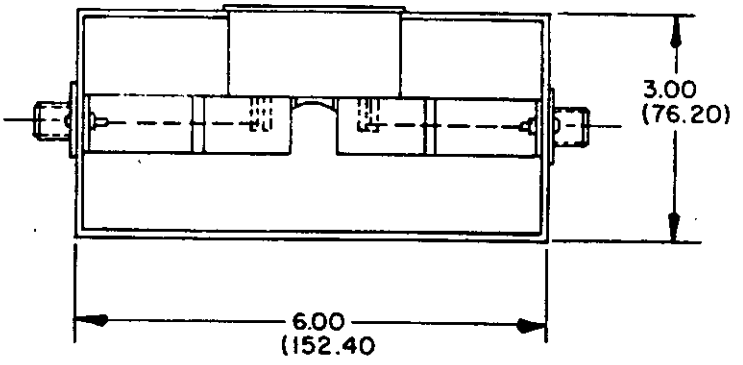
a. The axis Y – Y' is defined as the axis of the base pin gage described in note b.

b. The tube base shall be capable of entering to a distance of .375 (9.53 mm) a flat plate gage having six holes .0800±.0005 (2.03 ±.01 mm) and one hole .1450±.0005 (3.68±.01 mm) all arranged on a 1.0000±.0005 (25.40±.01 mm) circle at the angles ±5' specified on the outline. A .500 ±.010 (12.70±.25 mm) diameter hole at the center of the pin circle is also required. The axis Y – Y' is defined by the center of this hole.

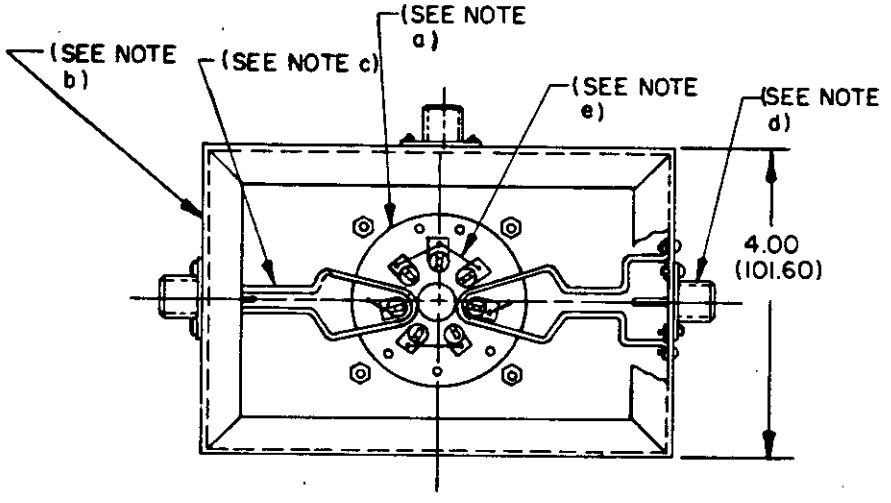
c. The plate leads shall be capable of entering a flat plate gage of .375±.001 (9.53±.03 mm) thickness having two holes .1420±.0005 (3.61±.01 mm) in diameter spaced .275±.001 (6.99±.03 mm) from a point coincident with the axis Y – Y'. The axis of the holes shall be parallel to Y – Y' and the plane of these axis shall be 90°±5' from the plane through Y – Y' and pin No. 4.

FIGURE 1. Outline drawing of electron tube types 6252 and 6907.

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SIDE VIEW

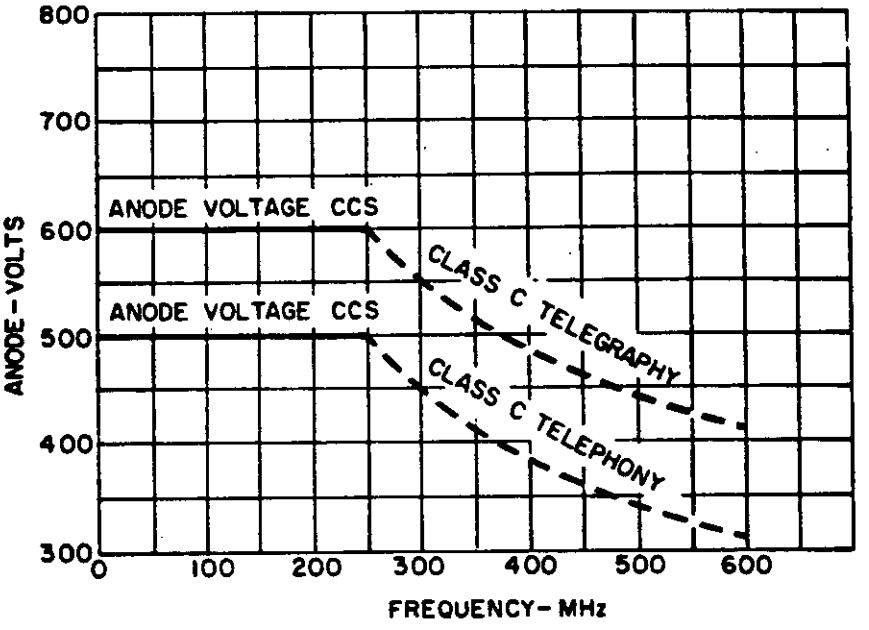


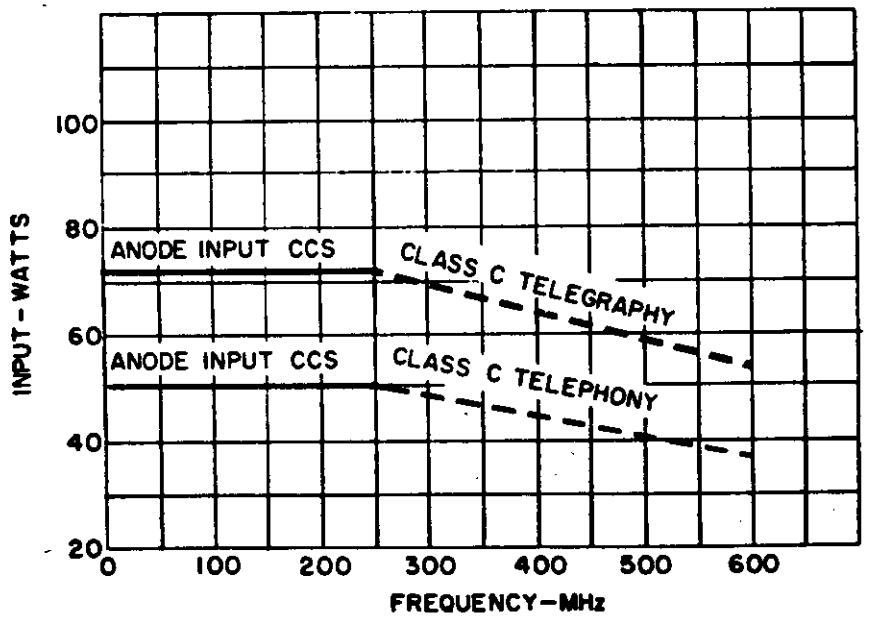
BOTTOM VIEW

NOTES:
a. 7-pin septar tube socket (Johnson No. 122-101-100 or equivalent).
b. Aluminum mounting box.
c. Copper shielding, .750 (19.05 mm) wide x .040 (1.02 mm) thick, may be hand formed to shape.
d. Female sockets for RF connectors, (Amphenol 50-239, or equivalent).
e. Copper connection wire .030 (.76 mm) diameter.
f. Dimensions are in inches.
g. Metric equivalents are in parentheses.

FIGURE 2. Mounting box with 7-pin septar socket, shielding and connection lead.

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NOTE:
Dotted curves apply to type 6907 only.

FIGURE 3. Input versus frequency curves.

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